

OpenStax Calculus Volume 2

Section 6.3 Exercises: Taylor and Maclaurin Series

Source: <https://openstax.org/books/calculus-volume-2/pages/6-3-taylor-and-maclaurin-series>. Exercises reproduced from OpenStax, licensed under CC BY 4.0.

This document was converted programmatically from OpenStax content; formulas were translated automatically from MathML and may contain conversion errors — verify against the source before relying on them.

In the following exercises, find the Taylor polynomials of degree two approximating the given function centered at the given point.

1. $f(x) = 1 + x + x^2$ at $a = 1$
2. $f(x) = 1 + x + x^2$ at $a = -1$
3. $f(x) = \cos(2x)$ at $a = \pi$
4. $f(x) = \sin(2x)$ at $a = \frac{\pi}{2}$
5. $f(x) = \sqrt{x}$ at $a = 4$
6. $f(x) = \ln x$ at $a = 1$
7. $f(x) = \frac{1}{x}$ at $a = 1$
8. $f(x) = e^x$ at $a = 1$

In the following exercises, verify that the given choice of n in the remainder estimate $|R_n| \leq \frac{M}{(n+1)!}(x-a)^{n+1}$, where M is the maximum value of $|f^{(n+1)}(z)|$ on the interval between a and the indicated point, yields $|R_n| \leq \frac{1}{1000}$. Find the value of the Taylor polynomial p_n of f at the indicated point. If $|R_n|$ is not less than $1/1000$, determine what it is.

9. [T] $\sqrt{10}$; $a = 9$, $n = 3$
10. [T] $28^{1/3}$; $a = 27$, $n = 1$
11. [T] $\sin(6)$; $a = 2\pi$, $n = 5$
12. [T] e^2 ; $a = 0$, $n = 9$
13. [T] $\cos\left(\frac{\pi}{5}\right)$; $a = 0$, $n = 4$
14. [T] $\ln(2)$; $a = 1$, $n = 1000$
15. Integrate the approximation $\sin t \approx t - \frac{t^3}{6} + \frac{t^5}{120} - \frac{t^7}{5040}$ evaluated at πt to approximate $\int_0^1 \frac{\sin(\pi t)}{\pi t} dt$.
16. Integrate the approximation $e^x \approx 1 + x + \frac{x^2}{2} + \cdots + \frac{x^6}{720}$ evaluated at $-x^2$ to approximate $\int_0^1 e^{-x^2} dx$.

In the following exercises, find the smallest value of n such that the remainder estimate $|R_n| \leq \frac{M}{(n+1)!}(x-a)^{n+1}$, where M is the maximum value of $|f^{(n+1)}(z)|$ on the interval between a and the indicated point, yields $|R_n| \leq \frac{1}{1000}$ on the indicated interval.

17. $f(x) = \sin x$ on $[-\pi, \pi]$, $a = 0$

18. $f(x) = \cos x$ on $[-\frac{\pi}{2}, \frac{\pi}{2}]$, $a = 0$

19. $f(x) = e^{-2x}$ on $[-1, 1]$, $a = 0$

20. $f(x) = e^{-x}$ on $[-3, 3]$, $a = 0$

In the following exercises, the maximum of the right-hand side of the remainder estimate $|R_1| \leq \frac{\max|f''(z)|}{2}R^2$ on $[a-R, a+R]$ occurs at a or $a \pm R$. Estimate the maximum value of R such that $\frac{\max|f''(z)|}{2}R^2 \leq 0.1$ on $[a-R, a+R]$ by plotting this maximum as a function of R .

21. [T] e^x approximated by $1+x$, $a = 0$

22. [T] $\sin x$ approximated by x , $a = 0$

23. [T] $\ln x$ approximated by $x-1$, $a = 1$

24. [T] $\cos x$ approximated by 1 , $a = 0$

In the following exercises, find the Taylor series of the given function centered at the indicated point.

25. x^4 at $a = -1$

26. $1+x+x^2+x^3$ at $a = -1$

27. $\sin x$ at $a = \pi$

28. $\cos x$ at $a = 2\pi$

29. $\sin x$ at $x = \frac{\pi}{2}$

30. $\cos x$ at $x = \frac{\pi}{2}$

31. e^x at $a = -1$

32. e^x at $a = 1$

33. $\frac{1}{(x-1)^2}$ at $a = 0$ (Hint: Differentiate $\frac{1}{1-x}$.)

34. $\frac{1}{(x-1)^3}$ at $a = 0$

35. $F(x) = \int_0^x \cos(\sqrt{t}) dt$; $f(t) = \sum_{n=0}^{\infty} (-1)^n \frac{t^n}{(2n)!}$ at $a = 0$ (Note: f is the Taylor series of $\cos(\sqrt{t})$.)

In the following exercises, compute the Taylor series of each function around $x = 1$.

36. $f(x) = 2 - x$

37. $f(x) = x^3$

38. $f(x) = (x-2)^2$

39. $f(x) = \ln x$

40. $f(x) = \frac{1}{x}$
41. $f(x) = \frac{1}{2x-x^2}$
42. $f(x) = \frac{x}{4x-2x^2-1}$
43. $f(x) = e^{-x}$
44. $f(x) = e^{2x}$

[T] In the following exercises, identify the value of x such that the given series $\sum_{n=0}^{\infty} a_n$ is the value of the Maclaurin series of f at x . Approximate $f(x)$ using $S_{10} = \sum_{n=0}^{10} a_n$.

45. $\sum_{n=0}^{\infty} \frac{1}{n!}$
46. $\sum_{n=0}^{\infty} \frac{2^n}{n!}$
47. $\sum_{n=0}^{\infty} \frac{(-1)^n (2\pi)^{2n}}{(2n)!}$
48. $\sum_{n=0}^{\infty} \frac{(-1)^n (2\pi)^{2n+1}}{(2n+1)!}$

The following exercises make use of the functions $S_5(x) = x - \frac{x^3}{6} + \frac{x^5}{120}$ and $C_4(x) = 1 - \frac{x^2}{2} + \frac{x^4}{24}$ on $[-\pi, \pi]$.

49. [T] Plot $\sin^2 x - (S_5(x))^2$ on $[-\pi, \pi]$. Compare the maximum difference with the square of the Taylor remainder estimate for $\sin x$.
50. [T] Plot $\cos^2 x - (C_4(x))^2$ on $[-\pi, \pi]$. Compare the maximum difference with the square of the Taylor remainder estimate for $\cos x$.
51. [T] Plot $|2S_5(x)C_4(x) - \sin(2x)|$ on $[-\pi, \pi]$.
52. [T] Compare $\frac{S_5(x)}{C_4(x)}$ on $[-1, 1]$ to $\tan x$. Compare this with the Taylor remainder estimate for the approximation of $\tan x$ by $x + \frac{x^3}{3} + \frac{2x^5}{15}$.
53. [T] Plot $e^x - e_4(x)$ where $e_4(x) = 1 + x + \frac{x^2}{2} + \frac{x^3}{6} + \frac{x^4}{24}$ on $[0, 2]$. Compare the maximum error with the Taylor remainder estimate.
54. (Taylor approximations and root finding.) Recall that Newton's method $x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$ approximates solutions of $f(x) = 0$ near the input x_0 .
 - (a) If f and g are inverse functions, explain why a solution of $g(x) = a$ is the value $f(a)$ of f .
 - (b) Let $p_N(x)$ be the degree- N Maclaurin polynomial of e^x . Use Newton's method to approximate solutions of $p_N(x) - 2 = 0$ for $N = 4, 5, 6$.
 - (c) Explain why the approximate roots of $p_N(x) - 2 = 0$ are approximate values of $\ln(2)$.

In the following exercises, use the fact that if $q(x) = \sum_{n=0}^{\infty} a_n(x-c)^n$ converges in an interval containing c , then $\lim_{x \rightarrow c} q(x) = a_0$ to evaluate each limit using Taylor series.

55. $\lim_{x \rightarrow 0} \frac{\cos x - 1}{x^2}$
56. $\lim_{x \rightarrow 0} \frac{\ln(1-x^2)}{x^2}$
57. $\lim_{x \rightarrow 0} \frac{e^{x^2} - x^2 - 1}{x^4}$

58. $\lim_{x \rightarrow 0^+} \frac{\cos(\sqrt{x}) - 1}{2x}$